Estimates of synthetic crude oil reserves are prepared by GLJ Petroleum Consultants of Calgary, Canada, third-party consultants. Their reports for all years are filed as exhibits to this Annual Report on Form 10-K. The team lead responsible for the estimates of our OSM reserves has 34 years of experience in petroleum engineering and has conducted surface mineable oil sands evaluations since 1986. He is a member of SPE, having served as regional director from 1998 through 2001. The second team member has 13 years of experience in petroleum engineering and has conducted surface mineable oil sands evaluations since 2009. Both are registered Practicing Professional Engineers in the Province of Alberta.

Audits of Estimates

Third-party consultants are engaged to provide independent estimates for fields that comprise 80 percent of our total proved reserves over a rolling four-year period for the purpose of auditing the in-house reserve estimates. We met this goal for the four-year period ended December 31, 2012. We established a tolerance level of 10 percent such that initial estimates by the third-party consultants are accepted if they are within 10 percent of our internal estimates. Should the third-party consultants' initial analysis fail to reach our tolerance level, both our team and the consultants re-examine the information provided, request additional data and refine their analysis if appropriate. This resolution process is continued until both estimates are within 10 percent. In the very limited instances where differences outside the 10 percent tolerance cannot be resolved by year end, a plan to resolve the difference is developed and our senior management is informed. This process did not result in significant changes to our reserve estimates in 2012 or 2011. There were no third-party audits performed in 2010.

During 2012, Netherland, Sewell & Associates, Inc. ("NSAI") prepared a Certification of December 31, 2011 reserves for the Alba field in E.G. The NSAI summary report is filed as an exhibit to this Annual Report on Form 10-K. Members of the NSAI team have many years of industry experience, having worked for large, international oil and gas companies before joining NSAI. The senior technical advisor has a Bachelor of Science degree in geophysics and over 15 years of experience in the estimation of and evaluation of reserves. The second member has a Bachelor of Science degree in chemical engineering and Master of Business Administration along with over 3 years of experience in estimation and evaluation of reserves. Both are licensed in the state of Texas.

Ryder Scott Company ("Ryder Scott") performed audits of several of our fields in 2012 and 2011. Their summary reports on audits performed in 2012 and 2011 are filed as exhibits to this Annual Report on Form 10-K. The team lead for Ryder Scott has over 20 years of industry experience, having worked for a major international oil and gas company before joining Ryder Scott. He has a Bachelor of Science degree in mechanical engineering, is a member of SPE where he served on the Oil and Gas Reserves Committee and is a registered Professional Engineer in the state of Texas.

Changes in Proved Undeveloped Reserves

As of December 31, 2012, 571 mmboe of proved undeveloped reserves were reported, an increase of 176 mmboe from December 31, 2011. The following table shows changes in total proved undeveloped reserves for 2012:

(mmboe)	
Beginning of year	395
Revisions of previous estimates	(13)
Improved recovery	2
Purchases of reserves in place	56
Extensions, discoveries, and other additions	201
Transfer to Proved Developed	(70)
End of year	571

Significant additions to proved undeveloped reserves during 2012 include 56 mmboe due to acquisitions in the Eagle Ford shale. Development drilling added 124 mmboe in the Eagle Ford, 35 mmboe in the Bakken and 15 mmboe in the Oklahoma Resource Basins shale play. A gas sharing agreement signed with the Libyan government in 2012 added 19 mmboe. Additionally, 30 mmboe were transferred from proved undeveloped to proved developed reserves in the Eagle Ford and 14 mmboe in the Bakken shale plays due to producing wells. Costs incurred in 2012, 2011 and 2010 relating to the development of proved undeveloped reserves, were $1,995 million $1,107 million and $1,463 million.

A total of 27 mmboe was booked as a result of reliable technology. Technologies included statistical analysis of production performance, decline curve analysis, rate transient analysis, reservoir simulation and volumetric analysis. The statistical nature of production performance coupled with highly certain reservoir continuity or quality within the reliable technology areas and sufficient proved undeveloped locations establish the reasonable certainty criteria required for booking reserves.